Fluency Skills by Lesson

The pre-lesson skills each lesson assumes — all 54 lessons, in teaching order

These are not the lesson's objectives. They are the earlier skills a student must produce quickly and correctly for the new teaching to land — the fluency a lesson spends none of its own time building. Each one carries its source: **"Gr 4"** means the skill is taught in Grade 4, **"2-3"** means this year's Lesson 2-3.

HOW TO USE THIS LIST

three minutes the day before you teach

1. Read the lesson's row the day before you teach it.
2. Put the quick check on the board for the first two minutes of class.
3. If most of the room misses it, the fluency gap — not the new content — is what will sink the lesson. The row names the earlier lesson to pull from.

THE FLUENCY SPINE

twelve skills, ordered by how many lessons depend on each

FLUENCY SKILL	LESSONS SERVED	WHY IT EARNS ITS SPOT
Recall multiplication and division facts through 12 × 12	3-1, 3-2, 3-3, 3-6, 5-1, 5-2, 5-5, 6-1, 6-3, 6-7, 8-3, 9-1	Slow facts turn every ratio table, area formula, and equation into decoding work.
Place a number on a scaled number line and read distance from it	2-1, 2-3, 2-4, 2-5, 2-6, 7-1, 7-2, 7-3, 7-4, 8-4, 8-5	All of statistics and all of Unit 7; one clothesline routine rehearses both.
Multiply and divide by 10, 100, and 1000 using place value	2-7, 3-6, 3-7, 4-1, 4-2, 4-3, 6-3	Makes metric conversion, percent, and benchmark estimation mental, not written.
Scale a pair of numbers by the same factor	3-3, 3-4, 3-5, 3-6, 3-7, 9-1, 9-3	The whole of Unit 3 and the table work in Unit 9 are this one move, repeated.
Plot and read ordered pairs on a coordinate plane	3-4, 7-5, 7-6, 7-7, 9-1, 9-2	Taught with ratios, reused in Unit 7, assumed in Unit 9; a weak 3-4 costs three units.
Order a list of numbers least to greatest	2-1, 2-3, 2-4, 2-5, 2-6, 4-2, 7-4	Needed for median, quartiles, IQR, and outliers. Cheap to rehearse, costly to skip.
Move among a fraction, a decimal, and a percent	4-1, 4-2, 4-3, 4-4, 4-5, 7-2, 7-4	All of Unit 4, plus placing rational numbers in Unit 7.
Follow order of operations, including parentheses and exponents	5-3, 6-4, 6-5, 6-6, 6-8, 9-4	Every formula and evaluation depends on it; trapezoid area is where it first bites.
Name the inverse of an operation and check by substituting back	4-5, 8-1, 8-2, 8-3, 9-4	The solving move in Unit 8, and the self-check that makes an answer trustworthy.
Add a list of numbers to a running total	2-2, 2-8, 2-9, 5-7, 5-8, 7-7	Means, MAD, surface area, and perimeter all end in one careful sum.
Find the area of a rectangle and of a triangle	5-1, 5-2, 5-3, 5-4, 5-6, 5-7, 5-8, 7-7	Every other figure in Unit 5 is decomposed into these two.
Multiply and divide with fractions and mixed numbers	4-4, 4-5, 5-2, 5-3, 5-5, 6-1, 6-2, 8-3	Grade 5 includes fraction multiplication and unit-fraction division; general fraction division is Grade 6 (6.NS.A.1).

UNIT 2

Statistics and Data Displays

Order a list of numbers, read a scaled number line, and add then divide a set of values. A student who cannot order numbers cannot find a median, a quartile, or an outlier.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
2-1 **Understand Statistical Questions** *Statistical Questions · Dot Plots and Shape*	**1** Order a list of up to 10 whole numbers least to greatest (Gr 4) **2** Read a scaled number line and say what one tick mark is worth (Gr 4) **3** Tally and count a set of up to 30 responses without losing place (Gr 3) **4** Compare two whole numbers and say which is greater (Gr 2)	Order 7, 3, 12, 5, 9. On a line from 0 to 10 split into 5 equal jumps, what is one jump worth?
2-2 **Represent and Describe Data in a Histogram** *Building a Histogram · Describing Histogram Shape*	**1** Decide which interval a value belongs to (is 19 in 10–19 or 20–29?) (Gr 5) **2** Sort up to 20 values into given intervals and count each bin (Gr 5) **3** Add a column of one- and two-digit numbers to a total (Gr 4) **4** Read a graph scale that skip-counts by 2, 5, or 10 (Gr 4)	Which interval holds 29: 20–29 or 30–39? How many of 4, 11, 19, 21, 25 land in 10–19?
2-3 **Describe the Data Using the Median** *Finding the Median · Median of an Even Set*	**1** Order up to 12 numbers least to greatest (Gr 4) **2** Find the middle position of an odd-length list by counting in from both ends (Gr 5) **3** Find the number halfway between two numbers (Gr 5) **4** Divide a two-digit sum by 2 (Gr 4)	Put in order: 9, 4, 15, 6, 11. What number is halfway between 12 and 17?
2-4 **Represent and Describe Data in a Box Plot** *Building a Box Plot · Comparing Box Plots*	**1** Order a data set and find its median (2-3) **2** Split an ordered list into a lower half and an upper half (2-3) **3** Subtract two-digit whole numbers (Gr 4) **4** Plot a point at the right place on a scaled number line (Gr 5)	Ordered: 3, 5, 8, 10, 12, 14, 18, 20. Find the median of the lower half.
2-5 **Describe Data by Range and Interquartile Range** *Range and IQR · Comparing Variability*	**1** Find the maximum and minimum of a data set (Gr 4) **2** Subtract whole numbers, including across zeros (Gr 4) **3** Locate Q1 and Q3 in an ordered set (2-4) **4** Compare two differences to say which spread is larger (Gr 4)	Range of 12, 7, 19, 4, 15? If Q1 = 6 and Q3 = 17, what is Q3 − Q1?
2-6 **Median and Outliers / Comparing Distributions** *Median and Outliers · Comparing Distributions*	**1** Find the median of an ordered set (2-3) **2** Recompute a median after one value is removed or changed (2-3) **3** Compare two numbers using < and > (Gr 3) **4** Judge whether one value sits far away from the rest of a set (Gr 5)	Median of 4, 6, 7, 8, 40? Now take out 40 — what is the median?

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
2-7 **Divide Decimals Using an Algorithm** *Dividing a Decimal by a Whole Number · Dividing by a Decimal*	**1** Recall division facts through 12 × 12 (Gr 3–4) **2** Divide multi-digit whole numbers with the standard algorithm (Gr 5) **3** Multiply and divide by 10, 100, 1000 using place value point (Gr 5) **4** Line up decimal places and annex zeros to keep dividing (Gr 5)	84 ÷ 4? 4.8 ÷ 2? 0.6 × 100?
2-8 **Describe Data Using the Mean** *Finding the Mean · Reaching a Target Mean*	**1** Add a list of 5–10 two-digit numbers accurately (Gr 4) **2** Divide a three-digit number by a one-digit number (Gr 5) **3** Count how many values are in a data set (Gr 3) **4** Find a missing addend that reaches a given total (Gr 4)	Mean of 10, 14, 8, 12, 16? What must be added to 48 to reach 60?
2-9 **Describe Data by Mean Absolute Deviation** *Finding the MAD · Comparing Sets with MAD*	**1** Find the mean of a data set (2-8) **2** Subtract to find the distance between two numbers and keep it positive (Gr 5) **3** Add a list of distances (Gr 4) **4** Divide a total by the number of values (2-8)	The mean is 20. How far is 26 from the mean? How far is 14?
2-10 **Choose Appropriate Measures** *Mean or Median? · Matching Center and Spread*	**1** Compute mean, median, range, IQR, and MAD on demand (2-3, 2-5, 2-8, 2-9) **2** Spot an outlier in a short list (2-6) **3** Read a dot plot or histogram as symmetric or skewed (2-1, 2-2) **4** Compare two measures and say which is pulled by an extreme value (2-8)	Which measure of center fits 2, 3, 3, 4, 50 — mean or median? Why?

UNIT 3

Ratios and Rates

Multiplication and division facts, and scaling a pair of numbers by the same factor. Every ratio table, graph, comparison, and conversion in this unit is that one move repeated.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
3-1 **Understand Ratios** *Understand Ratios · Ratios with Tape Diagrams*	**1** Recall multiplication facts through 12 × 12 (Gr 3) **2** Skip-count by any one-digit number (Gr 3) **3** Count each part of a part-whole picture, then add the parts for the total (Gr 2–4) **4** Write a fraction naming a part of a whole (Gr 4)	6 × 4? There are 3 dogs and 5 cats – how many animals in all?
3-2 **Rates and Unit Rates** *Understanding Unit Rates · Comparing Unit Prices (Best Buy)*	**1** Divide a multi-digit number by a one- or two-digit number (Gr 5) **2** Divide to find the amount for one, and read "per" as "for each one" (Gr 4) **3** Keep a repeating pattern going in a table (Gr 4) **4** Attach the correct unit label to a computed number (Gr 4)	60 ÷ 4? If 4 pencils cost $2, what does 1 pencil cost?
3-3 **Equivalent Ratios Using Tables** *Modeling Clay Recipe (ratio tables) · The Bridge Strategy (Scaling Ratios)*	**1** Multiply both numbers of a pair by the same factor (Gr 4) **2** Recall multiplication facts through 12 × 12 (Gr 3) **3** Write equivalent fractions by multiplying numerator and denominator (Gr 4) **4** Fill the next row of a growing pattern table (Gr 5)	2 × 5 and 3 × 5? Complete: 2 : 3 = 8 : ___
3-4 **Equivalent Ratios Using Graphs** *Graphing Ratios on Coordinate Planes · Interpreting Points on Ratio Graphs*	**1** Plot a point in Quadrant I from an ordered pair (Gr 5) **2** Read the scale on each axis before plotting (Gr 5) **3** Count right from the origin, then up (Gr 5) **4** Build a ratio table to generate the pairs (3-3)	Start at (0, 0), move 3 right and 2 up – write the ordered pair. If each gridline is 5, where is 15?
3-5 **Comparing Ratio Relationships** *Comparing Ratios with Common Terms · Comparing Rates with Benchmarks*	**1** Compare two whole numbers (Gr 2) **2** Scale both parts of a ratio to hit a target value (3-3) **3** Find a common multiple of two numbers (Gr 5) **4** Divide to find a unit rate for a fair comparison (3-2)	Which is greater, 12 or 9? Scale 3 : 4 so the first number is 12.
3-6 **Converting Within One System** *Customary Unit Conversions · Metric Unit Conversions*	**1** Recall multiplication and division facts (Gr 3) **2** Recall customary facts: 1 ft = 12 in, 1 lb = 16 oz, 1 gal = 4 qt (Gr 4–5) **3** Multiply and divide by 10, 100, 1000 for metric steps (Gr 5) **4** Decide whether a conversion calls for multiplying or dividing (Gr 5)	12 × 4? 1 foot = ___ inches? 2.5 m = ___ cm?

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
3-7 **Converting Between Systems** *Customary vs. Metric Conversions · Multi-Step System Conversions and Dosage*	**1** Multiply a whole number by a decimal (Gr 5) **2** Round a decimal to the nearest tenth or hundredth (Gr 5) **3** Multiply by a conversion factor and keep the unit label (3-6) **4** Chain two conversions in order without losing a step (3-6)	4 × 1.6? 10 × 0.45? Round 6.44 to the nearest tenth.

UNIT 4

Percent

Move fluently among a fraction, a decimal, and a percent, and multiply or divide by 100 using place value point. Benchmark percents (10%, 1%, 25%, 50%) do most of the work.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
4-1 **Understand Percent** *Percent Means Per Hundred · Percents Greater Than 100%*	**1** Read and write a fraction with denominator 100 (Gr 4) **2** Count shaded squares on a 100-square grid (Gr 4) **3** Write hundredths as a decimal (Gr 5) **4** Simplify a fraction over 100 (Gr 4)	37 of 100 squares are shaded – write it as a fraction, a decimal, and a percent.
4-2 **Relate Fractions, Decimals, and Percentages** *Fractions, Decimals, Percents · Compare and Order*	**1** Multiply and divide by 100 using place value point (Gr 5) **2** Write a fraction as a decimal by dividing (Gr 5) **3** Recall benchmark equivalents for 1/2, 1/4, 1/5, 3/4, 1/10 (Gr 4–5) **4** Compare decimals to the hundredths place (Gr 5)	0.45 as a percent? 25% as a fraction in simplest form? Which is greater, 0.6 or 55%?
4-3 **Estimate the Percent of a Number** *Estimate with Benchmarks · Estimate a Percent of a Number*	**1** Find 10% of a number using place value one place left (Gr 5) **2** Find 1% of a number by shifting two places left (Gr 5) **3** Halve and double a number mentally (Gr 4) **4** Round to a compatible number that is easy to take a percent of (Gr 5)	10% of 320? 1% of 320? Half of 32?
4-4 **Find and Compare with Percentages** *Find the Percent of a Number · Compare with Percents*	**1** Write a percent as a decimal (4-2) **2** Multiply a whole number by a decimal (Gr 5) **3** Multiply a whole number by a fraction (Gr 5) **4** Compare two products to decide which is larger (Gr 4)	0.25 × 80? 3/4 of 80? Which is more: 20% of 50 or 30% of 40?
4-5 **Determine the Whole Given the Part and Percent** *Find the Whole · Solve for the Whole*	**1** Divide a whole number by a decimal (2-7) **2** Tell which number is the part and which is the whole (4-4) **3** Divide by a benchmark fraction such as 1/4 or 1/2 (Gr 5) **4** Check an answer by multiplying back (Gr 4)	12 ÷ 0.25? 12 is 25% of what number? Now check: 25% of your answer.

UNIT 5

Area, Volume, and Surface Area

Multiply accurately, find the area of a rectangle and a triangle, and tell a perpendicular height from a slanted side. Every later formula in the unit is built from those two areas.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
5-1 **Determine the Area of Parallelograms and Rhombuses** *Area of Parallelograms · Area of Rhombuses*	**1** Multiply two- and three-digit numbers (Gr 5) **2** Find the area of a rectangle by multiplying length by width (Gr 3–4) **3** Tell a base from a slanted side and identify the perpendicular height (Gr 5) **4** Label an area answer in square units (Gr 4)	Area of an 8 cm by 5 cm rectangle? In a parallelogram, is the height the slanted side?
5-2 **Determine the Area of Triangles** *Area of Triangles · Missing Measures in Triangles*	**1** Multiply, then halve — or halve, then multiply (Gr 5) **2** Multiply a whole number by 1/2 (Gr 5) **3** Find the area of a rectangle or parallelogram (5-1) **4** Identify the height drawn perpendicular to the chosen base (5-1)	7 × 4, then half? Area of a triangle with base 7 m and height 4 m?
5-3 **Determine the Area of Trapezoids** *Trapezoids by Decomposing · The Trapezoid Area Formula*	**1** Add the two numbers inside parentheses before multiplying (Gr 5) **2** Multiply by 1/2 (5-2) **3** Follow order of operations inside a formula (Gr 5) **4** Split a figure into a rectangle plus triangles (5-2)	1/2 × (6 + 10) × 4 = ? Do the parentheses first.
5-4 **Apply Area Concepts to Solve Problems** *Area of Regular Polygons · Composite Figures*	**1** Find the area of rectangles, triangles, and trapezoids on demand (5-1, 5-2, 5-3) **2** Add and subtract areas (Gr 4) **3** Find a missing side length from the other labeled sides (Gr 5) **4** Multiply one part's area by the number of identical parts (Gr 4)	A 10 by 6 rectangle has a 3 by 2 rectangle cut out. What area is left?
5-5 **Determine the Volume of Rectangular Prisms** *Volume of Rectangular Prisms · Missing and Fractional Edges*	**1** Multiply three factors in any order (Gr 5) **2** Multiply fractions and mixed numbers (Gr 5) **3** Find the area of a rectangular base (5-1) **4** Label a volume answer in cubic units (Gr 5)	4 × 3 × 2? 1/2 × 1/3? Volume of a 4 by 3 by 2 prism?
5-6 **Represent Three-Dimensional Figures in Two Dimensions** *Drawing Nets of Solids · Reading Nets and Solids*	**1** Name the faces, edges, and vertices of a prism and a pyramid (Gr 5) **2** Count the faces of a solid (Gr 5) **3** Find the area of each rectangle and triangle inside a net (5-1, 5-2) **4** Match congruent faces in pairs (Gr 4)	How many faces does a rectangular prism have? How many matching pairs is that?

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
5-7 **Determine Surface Area of Prisms** *Surface Area of Rectangular Prisms · Surface Area of Triangular Prisms*	**1** Add a list of five or six areas (Gr 4) **2** Find the area of rectangles and triangles quickly (5-1, 5-2) **3** Read a net and pair up matching faces (5-6) **4** Double one face's area to cover a matching pair (Gr 4)	Faces measure 12, 12, 20, 20, 15, 15 square units – what is the surface area?
5-8 **Determine Surface Area of Pyramids** *Surface Area of Pyramids · Lateral Area of Pyramids*	**1** Find the area of a triangle (5-2) **2** Find the area of a square or rectangular base (5-1) **3** Multiply one triangular face's area by the number of identical faces (Gr 4) **4** Tell slant height from vertical height on a net or drawing (5-6)	One triangular face is 24 square inches. With four identical faces, what is the lateral area?

UNIT 6

Expressions

Order of operations, and the distributive property in both directions. Fraction division opens the unit, so reciprocals and improper fractions must be automatic first.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
6-1 **Division Expressions with Fractions and Whole Numbers** *Dividing by a Unit Fraction · Dividing a Fraction by a Whole Number*	**1** Divide decimals and whole numbers (2-7) **2** Name the reciprocal of a whole number and of a unit fraction (Gr 5) **3** Multiply a whole number by a fraction (Gr 5) **4** Read a fraction bar as "divided by" (Gr 5)	4.8 ÷ 2? How many fourths are in 3? Reciprocal of 5?
6-2 **Division Expressions with Fractions and Mixed Numbers** *Dividing a Fraction by a Fraction · Dividing with Mixed Numbers*	**1** Rewrite a mixed number as an improper fraction (Gr 5) **2** Multiply two fractions (Gr 5) **3** Find the reciprocal of any fraction (6-1) **4** Simplify a fraction to lowest terms (Gr 4)	2 3/4 as an improper fraction? Reciprocal of 3/5? 2/3 × 5/4?
6-3 **Explore Numerical Expressions with Exponents** *Writing Repeated Factors as Powers · Finding the Value of a Power*	**1** Multiply the same factor repeatedly (Gr 4) **2** Recall 2^2, 3^2, 4^2, 5^2, 2^3, and 3^3 (Gr 5–6) **3** Multiply by 10 repeatedly to build powers of 10 (Gr 5) **4** Count how many factors are written (Gr 3)	6^2? Write 2 × 2 × 2 × 2 as a power. 10^3?
6-4 **Write and Evaluate Numerical Expressions with Exponents** *Writing Numerical Expressions · Order of Operations with Powers*	**1** Evaluate a power (6-3) **2** Follow order of operations with parentheses and exponents (Gr 5) **3** Read a fraction bar as a grouping symbol (Gr 5) **4** Translate "sum," "product," and "less than" into symbols (Gr 5)	4 + 6 ÷ 2? 3 + 2^2 × 5? Which operation comes first?
6-5 **Write and Evaluate Algebraic Expressions** *Writing Algebraic Expressions · Evaluating Algebraic Expressions*	**1** Substitute a number for a letter, then compute (Gr 6) **2** Follow order of operations after substituting (6-4) **3** Read 4n as 4 × n (Gr 6) **4** Translate a word phrase into an expression (6-4)	Evaluate 4n + 3 when n = 5. Write an expression for "7 more than a number."

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
6-6 **Identify Equivalent Algebraic Expressions** *Simplifying to Show Equivalence · Testing and Justifying Equivalence*	**1** Combine like terms such as 5x + 2x (Gr 6) **2** Evaluate an expression at a chosen value (6-5) **3** Apply the distributive property with numbers, e.g. 3(4 + 5) (Gr 5) **4** Compare two results to decide whether expressions match (Gr 4)	5x + 2x = ? Evaluate 2(x + 3) and 2x + 6 at x = 4 — are they the same?
6-7 **Find Factors and Multiples** *Factors and the Greatest Common Factor · Multiples and the Least Common Multiple*	**1** Recall multiplication and division facts (Gr 3) **2** List every factor pair of a number up to 100 (Gr 4) **3** List the first several multiples of a number (Gr 4) **4** Test divisibility by 2, 3, 5, 9, and 10 (Gr 5)	All the factors of 18? First five multiples of 6? GCF of 18 and 24?
6-8 **Generate Equivalent Expressions** *Using Properties to Rewrite Expressions · Expanding and Factoring Expressions*	**1** Find the GCF of two numbers (6-7) **2** Apply the distributive property forward and backward (6-6) **3** Combine like terms (6-6) **4** Multiply a number by a sum (Gr 5)	GCF of 18 and 24? Expand 3(x + 5). Factor 4x + 12.

UNIT 7

Integers, Rational Numbers, and the Coordinate Plane

Place any number on a number line and read distance from it. Absolute value, comparison, and coordinate distance are all the same number-line move.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
7-1 **Explore Integers and Their Opposites** *Integers and Their Opposites · Integers in Real Situations*	**1** Place whole numbers on a number line (Gr 4) **2** Count forward and backward through zero (Gr 5) **3** Read a negative sign as "below zero" or "the opposite of" (Gr 6) **4** Simplify an algebraic expression carried in from Unit 6 (6-6)	What is the opposite of −6? Where does −3 sit compared with 0?
7-2 **Represent Rational Numbers and Their Opposites on the Number Line** *Rational Numbers on a Number Line · Opposites of Rational Numbers*	**1** Place fractions and decimals between whole numbers (Gr 5) **2** Name the two consecutive integers a number falls between (Gr 5) **3** Convert between a fraction and a decimal (4-2) **4** Partition a number-line interval into equal parts (Gr 4)	Which two integers does 2.7 fall between? Where does −11/2 go?
7-3 **Understand Absolute Value of Rational Numbers** *Absolute Value as Distance · Absolute Value and Total Distance*	**1** Count the distance between two points on a number line (Gr 4) **2** Name the opposite of a number (7-1) **3** Report a distance as a positive number (Gr 6) **4** Add two distances to find a total (Gr 3)	|−9|? How far is −4 from 0? How far from −4 to 3?
7-4 **Compare and Order Integers and Rational Numbers** *Comparing Numbers on a Number Line · Ordering Integers and Rational Numbers*	**1** Use < and > correctly (Gr 3) **2** Place numbers on a number line (7-2) **3** Compare decimals and fractions (4-2) **4** Read left-is-less, right-is-greater on a number line (7-1)	−6 ___ −2. Order these: −3, 1.5, −1/2, 0.
7-5 **Represent Rational Numbers on the Coordinate Plane** *Plotting Points on the Coordinate Plane · Reflecting Points Across the Axes*	**1** Plot ordered pairs in Quadrant I (3-4) **2** Read the sign of each coordinate to name a quadrant (Gr 6) **3** Move left or right first, then up or down from the origin (Gr 5) **4** Read a scaled axis before plotting (3-4)	Which quadrant holds (−4, 5)? Plot (3, −2).
7-6 **Determine Distance on the Coordinate Plane** *Distance Between Points on a Grid · Comparing Distances on the Plane*	**1** Subtract integers on a number line (7-4) **2** Use absolute value to report a distance (7-3) **3** Read the coordinates off a plotted point (7-5) **4** Add two distances when a segment crosses an axis (7-3)	Distance from (−5, 2) to (3, 2)? What is |−5 − 3|?

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
7-7 **Represent Polygons on the Coordinate Plane** *Polygons and Perimeter on the Plane · Designing Polygons with a Scale*	**1** Plot several points and connect them in order (7-5) **2** Find horizontal and vertical distances between vertices (7-6) **3** Add side lengths to find perimeter (Gr 4) **4** Find the area of a rectangle or triangle (5-1, 5-2)	Plot (1,1), (5,1), (5,4), (1,4). What is the perimeter? The area?

UNIT 8

Equations and Inequalities

Inverse operations, and substituting a value back in to check. Students who do not know what undoes an operation cannot solve or verify anything in this unit.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
8-1 **Understand Equations and Their Solutions** *Equations and Solutions · Writing Equations*	**1** Substitute a value into an expression and evaluate (6-5) **2** Read an equals sign as a balance, not as "the answer is" (Gr 4) **3** Translate a sentence into an equation (6-5) **4** Compare two sides to decide whether they match (6-6)	Is x = 4 a solution of 3x = 12? Write an equation for "three times a number is 15."
8-2 **Write and Solve Equations Using Addition or Subtraction** *Addition Equations · Subtraction Equations*	**1** Recall addition and subtraction facts and their inverses (Gr 3) **2** Add and subtract multi-digit numbers, decimals, and fractions (Gr 5) **3** Name the inverse of adding or subtracting a number (Gr 4) **4** Substitute the answer back in to check (8-1)	m − 11 = 25, so m = ? What undoes "+ 8"?
8-3 **Write and Solve Equations Using Multiplication or Division** *Multiplication Equations · Division Equations*	**1** Recall multiplication and division facts (Gr 3) **2** Divide multi-digit numbers and decimals (2-7) **3** Name the inverse of multiplying or dividing by a number (Gr 4) **4** Divide by a fraction to undo multiplying by one (6-2)	6x = 42, so x = ? x ÷ 5 = 9, so x = ? What undoes "× 4"?
8-4 **Write and Represent Inequalities** *Writing Inequalities · Representing Inequalities*	**1** Use <, >, ≤, and ≥ correctly (7-4) **2** Translate "at least," "at most," "no more than," "more than" (Gr 6) **3** Place a number on a number line (7-2) **4** Decide whether a given value makes a comparison true (8-1)	Write an inequality for "you must be at least 48 inches tall." Is 50 a solution?
8-5 **Understand Inequalities and Their Solutions** *Graphing Inequalities · Testing Solutions*	**1** Mark a boundary point on a number line and shade one direction (8-4) **2** Choose an open or closed circle from the symbol (8-4) **3** Substitute and test several candidate values (8-1) **4** Compare a value against the boundary number (7-4)	Graph x ≥ 3. Open circle or closed? Is 3 itself a solution?

UNIT 9

Relationships Between Two Variables

The unit rate from Unit 3, the table-to-graph move from Unit 3, and substitution from Units 6 and 8. Unit 9 asks students to carry one relationship through a table, a graph, and an equation.

LESSON	FLUENCY SKILLS NEEDED FIRST	QUICK CHECK
9-1 **Explore Relationships Between Two Variables** *Independent and Dependent Variables · Tables of Values*	**1** Evaluate an expression for several input values (6-5) **2** Build a table of values from a rule (3-3) **3** Multiply a unit rate by a quantity (3-2) **4** Tell which quantity is chosen and which one responds (Gr 6)	Songs cost $2 each. Which quantity is independent? Complete the table for 1, 2, and 3 songs.
9-2 **Analyze Graphs of Relationships Between Two Variables** *From Table to Graph · Analyzing a Graph*	**1** Plot ordered pairs taken from a table (3-4, 7-5) **2** Read a point's coordinates off a graph (7-5) **3** Read the scale on each axis (3-4) **4** Extend a pattern to predict the next point (3-3)	In (x, y), which value comes first? Read the point at 4 across and 12 up.
9-3 **Write Equations to Represent Relationships Between Two Variables** *Equations from Tables · Equations from Situations*	**1** Find the unit rate from a table (3-2) **2** Divide y by x to find the constant factor (3-2) **3** Write an expression using a variable (6-5) **4** Test an equation against every row of the table (8-1)	Table: (1, 3), (2, 6), (3, 9). Write y in terms of x.
9-4 **Apply Two-Variable Relationships to Solve Problems** *Substitute and Solve · Working Backward to a Goal*	**1** Substitute into an equation and evaluate (6-5) **2** Solve a one-step equation for the other variable (8-2, 8-3) **3** Multiply and divide with decimals (2-7) **4** Read the answer back in the situation's units (3-2)	Muffins cost $3 each, so y = 3x. How many muffins can you buy with $24?